```
!pip install -q pandas numpy matplotlib seaborn scikit-learn openpyxl
```

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

print("Customer Segmentation Environment Ready!")
```

```
Customer Segmentation Environment Ready!
```

```
from google.colab import files

uploaded = files.upload()
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving customer_segmentation.csv to customer_segmentation (1).csv

```
df = pd.read_csv("customer_segmentation.csv")

print("Dataset loaded successfully!")
print("Rows and Columns:", df.shape)

display(df.head())
```

```
Dataset loaded successfully!
Rows and Columns: (2240, 29)
```

	ID	Year_Birth	Education	Marital_Status	Income	Kidhome	Teenhome	Dt_Customer	Recency	MntWines	...	NumWebVisits
0	5524	1957	Graduation	Single	58138.0	0	0	04-09-2012	58	635	...	
1	2174	1954	Graduation	Single	46344.0	1	1	08-03-2014	38	11	...	
2	4141	1965	Graduation	Together	71613.0	0	0	21-08-2013	26	426	...	
3	6182	1984	Graduation	Together	26646.0	1	0	10-02-2014	26	11	...	
4	5324	1981	PhD	Married	58293.0	1	0	19-01-2014	94	173	...	

5 rows × 29 columns

```
df.describe()
```

	ID	Year_Birth	Income	Kidhome	Teenhome	Recency	MntWines	MntFruits	MntMeatProduct
count	2240.000000	2240.000000	2216.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.00000
mean	5592.159821	1968.805804	52247.251354	0.444196	0.506250	49.109375	303.935714	26.302232	166.95000
std	3246.662198	11.984069	25173.076661	0.538398	0.544538	28.962453	336.597393	39.773434	225.71537
min	0.000000	1893.000000	1730.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.00000
25%	2828.250000	1959.000000	35303.000000	0.000000	0.000000	24.000000	23.750000	1.000000	16.00000
50%	5458.500000	1970.000000	51381.500000	0.000000	0.000000	49.000000	173.500000	8.000000	67.00000
75%	8427.750000	1977.000000	68522.000000	1.000000	1.000000	74.000000	504.250000	33.000000	232.00000
max	11191.000000	1996.000000	666666.000000	2.000000	2.000000	99.000000	1493.000000	199.000000	1725.00000

8 rows × 26 columns

```
df.columns
```

```
Index(['ID', 'Year_Birth', 'Education', 'Marital_Status', 'Income', 'Kidhome',
       'Teenhome', 'Dt_Customer', 'Recency', 'MntWines', 'MntFruits',
       'MntMeatProducts', 'MntFishProducts', 'MntSweetProducts',
       'MntGoldProds', 'NumDealsPurchases', 'NumWebPurchases',
       'NumCatalogPurchases', 'NumStorePurchases', 'NumWebVisitsMonth',
       'AcceptedCmp3', 'AcceptedCmp4', 'AcceptedCmp5', 'AcceptedCmp1',
       'AcceptedCmp2', 'Complain', 'Z_CostContact', 'Z_Revenue', 'Response'],
      dtype='object')
```

```
df.shape
```

```
(2240, 29)
```

```
df.isna().sum()
```

	0
ID	0
Year_Birth	0
Education	0
Marital_Status	0
Income	24
Kidhome	0
Teenhome	0
Dt_Customer	0
Recency	0
MntWines	0
MntFruits	0
MntMeatProducts	0
MntFishProducts	0
MntSweetProducts	0
MntGoldProds	0
NumDealsPurchases	0
NumWebPurchases	0
NumCatalogPurchases	0
NumStorePurchases	0
NumWebVisitsMonth	0
AcceptedCmp3	0
AcceptedCmp4	0
AcceptedCmp5	0
AcceptedCmp1	0
AcceptedCmp2	0
Complain	0
Z_CostContact	0
Z_Revenue	0
Response	0

dtype: int64

```
df.isna().sum().sum()
```

```
np.int64(24)
```

```
df.dropna(inplace=True)
```

```
df.isna().sum().sum()
```

```
np.int64(0)
```

```
df.describe()
```

	ID	Year_Birth	Income	Kidhome	Teenhome	Recency	MntWines	MntFruits	MntMeatProduct
count	2216.000000	2216.000000	2216.000000	2216.000000	2216.000000	2216.000000	2216.000000	2216.000000	2216.00000
mean	5588.353339	1968.820397	52247.251354	0.441787	0.505415	49.012635	305.091606	26.356047	166.99593
std	3249.376275	11.985554	25173.076661	0.536896	0.544181	28.948352	337.327920	39.793917	224.28327
min	0.000000	1893.000000	1730.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.00000
25%	2814.750000	1959.000000	35303.000000	0.000000	0.000000	24.000000	24.000000	2.000000	16.00000
50%	5458.500000	1970.000000	51381.500000	0.000000	0.000000	49.000000	174.500000	8.000000	68.00000
75%	8421.750000	1977.000000	68522.000000	1.000000	1.000000	74.000000	505.000000	33.000000	232.25000
max	11191.000000	1996.000000	666666.000000	2.000000	2.000000	99.000000	1493.000000	199.000000	1725.00000

8 rows × 26 columns

```
df['Education'].value_counts()
```

Education	count
Graduation	1116
PhD	481
Master	365
2n Cycle	200
Basic	54

dtype: int64

```
df['Marital_Status'].value_counts()
```

Marital_Status	count
Married	857
Together	573
Single	471
Divorced	232
Widow	76
Alone	3
Absurd	2
YOLO	2

dtype: int64

DATA CLEANING AND FEATURE ENGINEERING

```
df["Dt_Customer"]=pd.to_datetime(df["Dt_Customer"],dayfirst=True)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 2216 entries, 0 to 2239
Data columns (total 29 columns):
 #   Column               Non-Null Count  Dtype         
---  ------               --------------  -----         
 0   ID                   2216 non-null   int64         
 1   Year_Birth           2216 non-null   int64         
 2   Education            2216 non-null   object        
 3   Marital_Status       2216 non-null   object        
 4   Income               2216 non-null   float64       
 5   Kidhome              2216 non-null   int64         
 6   Teenhome             2216 non-null   int64         
 7   Dt_Customer          2216 non-null   datetime64[ns]
 8   Recency              2216 non-null   int64         
 9   MntWines             2216 non-null   int64         
 10  MntFruits            2216 non-null   int64         
 11  MntMeatProducts      2216 non-null   int64         
 12  MntFishProducts      2216 non-null   int64         
```

```
 13  MntSweetProducts     2216 non-null   int64
 14  MntGoldProds         2216 non-null   int64
 15  NumDealsPurchases    2216 non-null   int64
 16  NumWebPurchases      2216 non-null   int64
 17  NumCatalogPurchases  2216 non-null   int64
 18  NumStorePurchases    2216 non-null   int64
 19  NumWebVisitsMonth    2216 non-null   int64
 20  AcceptedCmp3         2216 non-null   int64
 21  AcceptedCmp4         2216 non-null   int64
 22  AcceptedCmp5         2216 non-null   int64
 23  AcceptedCmp1         2216 non-null   int64
 24  AcceptedCmp2         2216 non-null   int64
 25  Complain             2216 non-null   int64
 26  Z_CostContact        2216 non-null   int64
 27  Z_Revenue            2216 non-null   int64
 28  Response             2216 non-null   int64
dtypes: datetime64[ns](1), float64(1), int64(25), object(2)
memory usage: 519.4+ KB
```

Double-click (or enter) to edit

```
# Calculate Age using Year_Birth (using 2025 as the baseline year)
df["Age"] = 2025 - df["Year_Birth"]
```

```
df["Age"]
```

	Age
0	68
1	71
2	60
3	41
4	44
...	...
2235	58
2236	79
2237	44
2238	69
2239	71

2216 rows × 1 columns

dtype: int64

```
df.head()
```

	ID	Year_Birth	Education	Marital_Status	Income	Kidhome	Teenhome	Dt_Customer	Recency	MntWines	...	AcceptedCmp3
0	5524	1957	Graduation	Single	58138.0	0	0	2012-09-04	58	635	...	0
1	2174	1954	Graduation	Single	46344.0	1	1	2014-03-08	38	11	...	0
2	4141	1965	Graduation	Together	71613.0	0	0	2013-08-21	26	426	...	0
3	6182	1984	Graduation	Together	26646.0	1	0	2014-02-10	26	11	...	0
4	5324	1981	PhD	Married	58293.0	1	0	2014-01-19	94	173	...	0

5 rows × 30 columns

```
df["Total_Children"] = df["Kidhome"] + df["Teenhome"]
df["Total_Children"]
```

	Total_Children
0	0
1	2
2	0
3	1
4	1
...	...
2235	1
2236	3
2237	0
2238	1
2239	2

2216 rows × 1 columns

dtype: int64

```
df.columns
```

```
Index(['ID', 'Year_Birth', 'Education', 'Marital_Status', 'Income', 'Kidhome',
       'Teenhome', 'Dt_Customer', 'Recency', 'MntWines', 'MntFruits',
       'MntMeatProducts', 'MntFishProducts', 'MntSweetProducts',
       'MntGoldProds', 'NumDealsPurchases', 'NumWebPurchases',
       'NumCatalogPurchases', 'NumStorePurchases', 'NumWebVisitsMonth',
       'AcceptedCmp3', 'AcceptedCmp4', 'AcceptedCmp5', 'AcceptedCmp1',
       'AcceptedCmp2', 'Complain', 'Z_CostContact', 'Z_Revenue', 'Response',
       'Age', 'Total_Children'],
      dtype='object')
```

```
spend_cols = ['MntWines', 'MntFruits', 'MntMeatProducts', 'MntFishProducts', 'MntSweetProducts', 'MntGoldProds']
df["Total_Spend"] = df[spend_cols].sum(axis=1)
```

```
df["Total_Spend"]
```

	Total_Spend
0	1617
1	27
2	776
3	53
4	422
...	...
2235	1341
2236	444
2237	1241
2238	843
2239	172

2216 rows × 1 columns

dtype: int64

```
df["Customer_Since"] = (pd.Timestamp("today") - df["Dt_Customer"]).dt.days
```

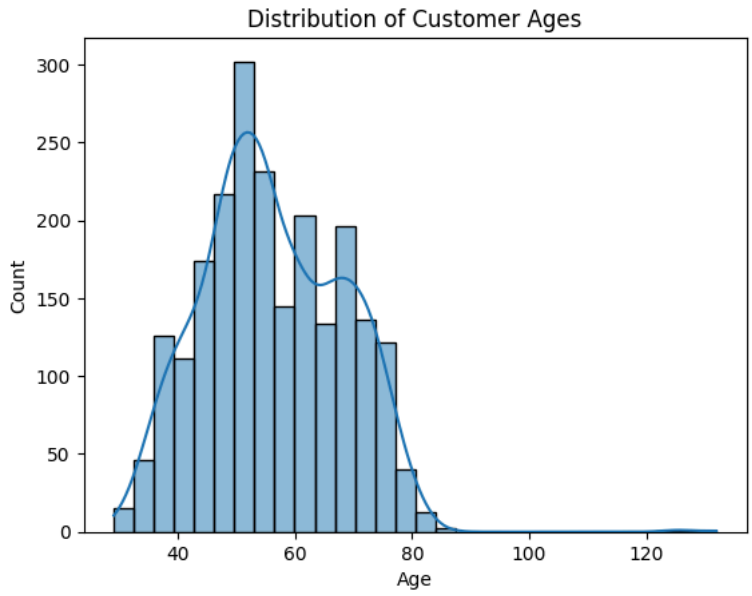
```
df["Customer_Since"]
```

	Customer_Since
0	5173
1	4623
2	4822
3	4649
4	4671
...	...
2235	4891
2236	4529
2237	4665
2238	4666
2239	5132

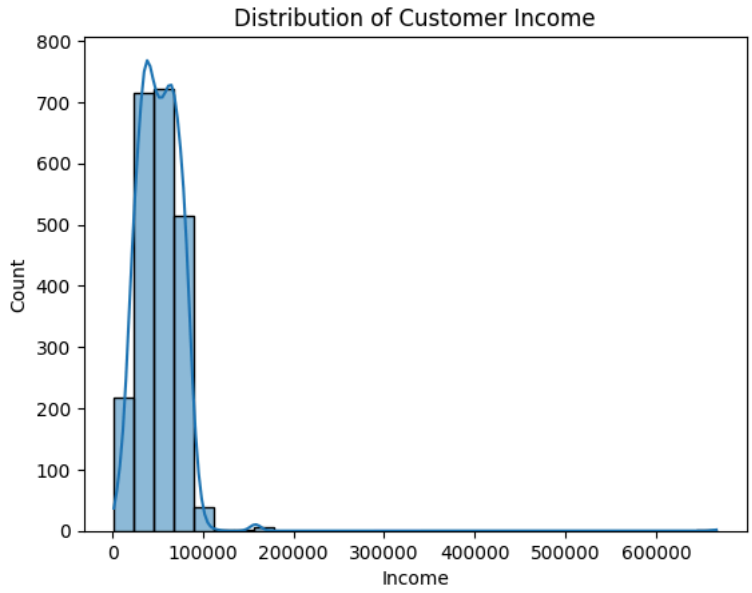
2216 rows × 1 columns

dtype: int64

```
sns.histplot(df["Age"], bins=30, kde=True)
plt.title("Distribution of Customer Ages")
plt.xlabel("Age")
plt.ylabel("Count")
plt.show()
```




```
sns.histplot(df["Income"],bins=30,kde=True)
plt.title("Distribution of Customer Income")
plt.xlabel("Income")
plt.ylabel("Count")
plt.show()
```

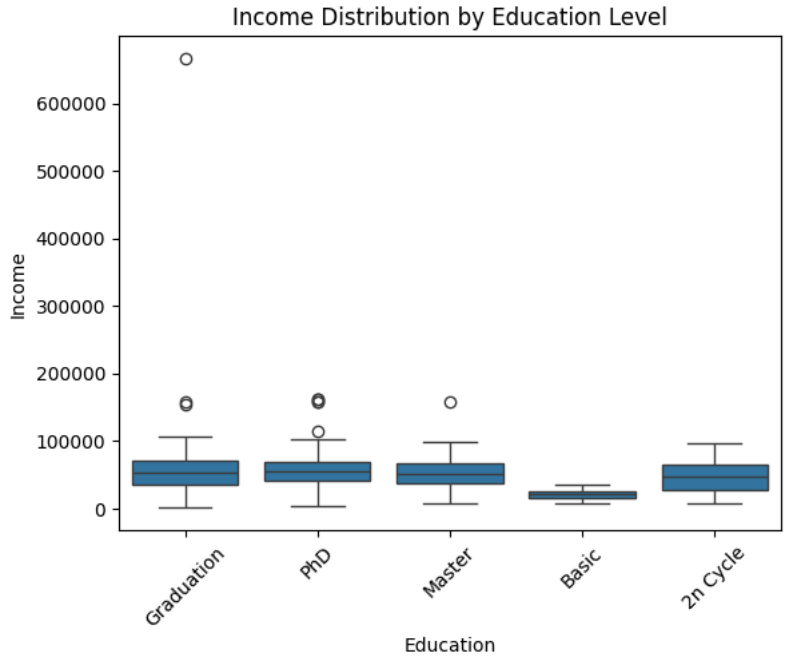



```
sns.histplot(df["Total_Spend"], bins=30, kde=True)
plt.title("Distribution of Customer Spending")
plt.xlabel("Spending")
plt.ylabel("Count")
plt.show()
```

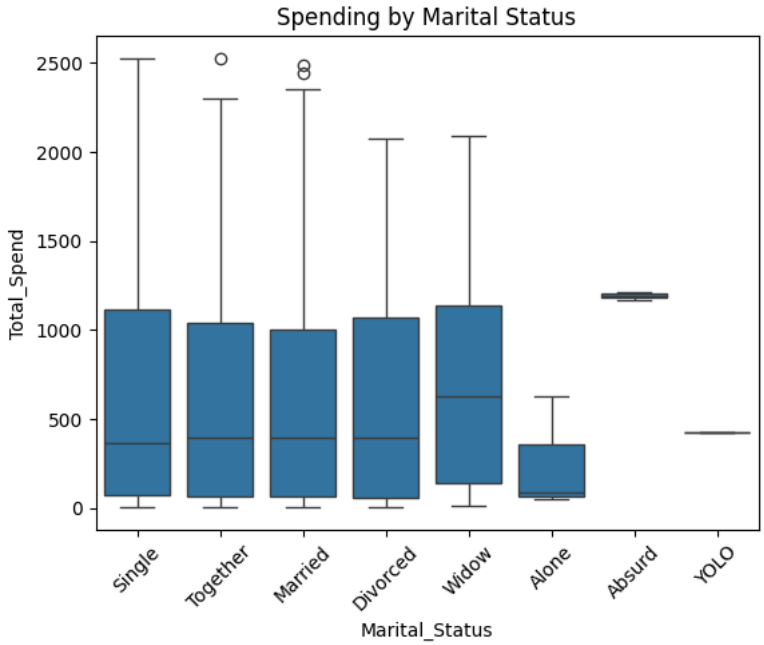



```
sns.boxplot(x="Education",y="Income", data=df)
plt.xticks(rotation=45)
plt.title("Income Distribution by Education Level")
plt.xlabel("Education")
plt.ylabel("Income")
```

```
Text(0, 0.5, 'Income')
```




```
sns.boxplot(x="Marital_Status",y="Total_Spend", data=df)
plt.xticks(rotation=45)
plt.title("Spending by Marital Status")
plt.show()
```




```
df.columns
```

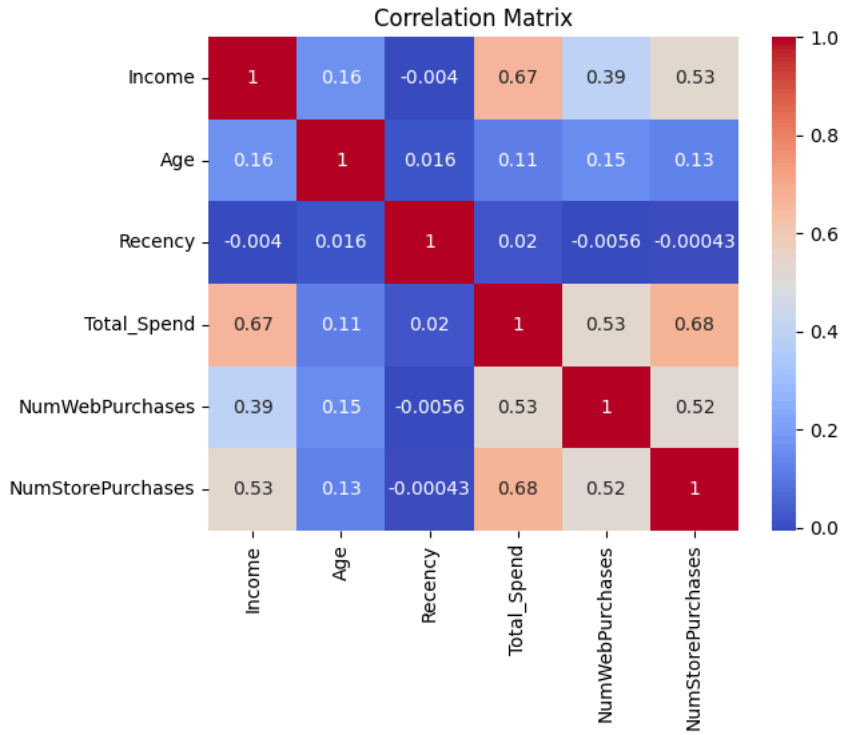
```
Index(['ID', 'Year_Birth', 'Education', 'Marital_Status', 'Income', 'Kidhome',
       'Teenhome', 'Dt_Customer', 'Recency', 'MntWines', 'MntFruits',
       'MntMeatProducts', 'MntFishProducts', 'MntSweetProducts',
       'MntGoldProds', 'NumDealsPurchases', 'NumWebPurchases',
       'NumCatalogPurchases', 'NumStorePurchases', 'NumWebVisitsMonth',
       'AcceptedCmp3', 'AcceptedCmp4', 'AcceptedCmp5', 'AcceptedCmp1',
       'AcceptedCmp2', 'Complain', 'Z_CostContact', 'Z_Revenue', 'Response',
       'Age', 'Total_Children', 'Total_Spend', 'Customer_Since'],
      dtype='object')
```

Start coding or generate with AI.

```
corr = df[["Income", "Age", "Recency", "Total_Spend", "NumWebPurchases", "NumStorePurchases"]].corr()
corr
```

	Income	Age	Recency	Total_Spend	NumWebPurchases	NumStorePurchases
Income	1.000000	0.161791	-0.003970	0.667576	0.387878	0.529362
Age	0.161791	1.000000	0.016295	0.113487	0.153051	0.127891
Recency	-0.003970	0.016295	1.000000	0.020066	-0.005641	-0.000434
Total_Spend	0.667576	0.113487	0.020066	1.000000	0.528973	0.675181
NumWebPurchases	0.387878	0.153051	-0.005641	0.528973	1.000000	0.516240
NumStorePurchases	0.529362	0.127891	-0.000434	0.675181	0.516240	1.000000

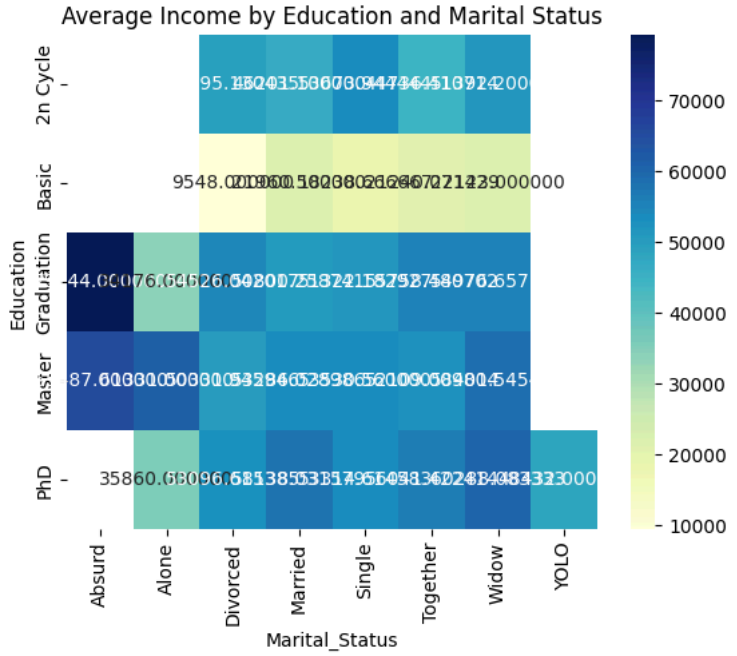
```
sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.title("Correlation Matrix")
plt.show()
```




```
pivot_income = df.pivot_table(values="Income", index="Education", columns="Marital_Status", aggfunc="mean")
pivot_income
```

Marital_Status Education	Absurd	Alone	Divorced	Married	Single	Together	Widow	YOLO
2n Cycle	NaN	NaN	49395.130435	46201.100000	53673.944444	44736.410714	51392.200000	NaN
Basic	NaN	NaN	9548.000000	21960.500000	18238.666667	21240.071429	22123.000000	NaN
Graduation	79244.0	34176.0	54526.042017	50800.258741	51322.182927	55758.480702	54976.657143	NaN
Master	65487.0	61331.0	50331.945946	53286.028986	53530.560000	52109.009804	58401.545455	NaN
PhD	NaN	35860.0	53096.615385	58138.031579	53314.614583	56041.422414	60288.083333	48432.0

```
sns.heatmap(pivot_income, annot=True, fmt="0f", cmap="YlGnBu")
plt.title("Average Income by Education and Marital Status")
plt.show()
```



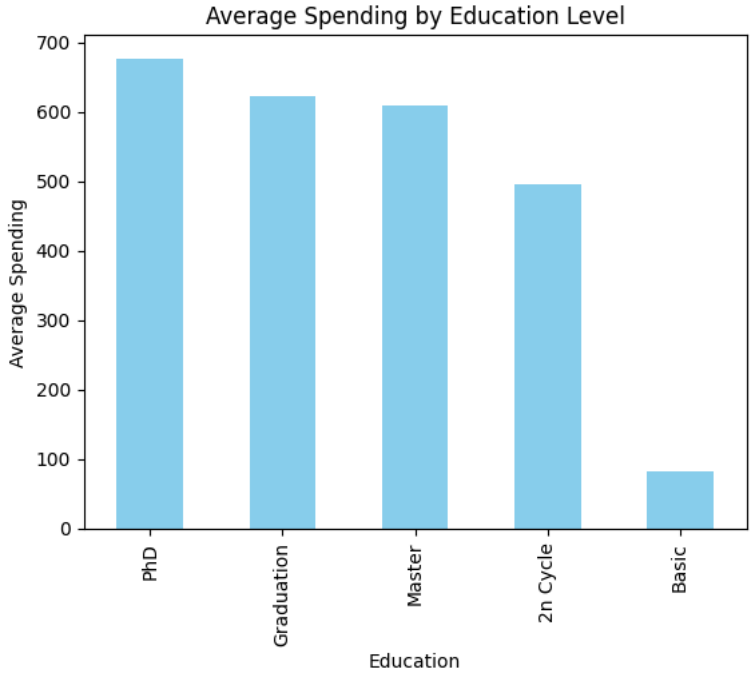

```
group1 = df.groupby("Education")["Total_Spend"].mean().sort_values(ascending=False)
group1
```

Education	Total_Spend
PhD	676.733888
Graduation	621.686380
Master	609.767123
2n Cycle	494.930000
Basic	81.796296

dtype: float64

```
group1.plot(kind="bar",
color="skyblue", title="Average Spending by Education Level")
plt.xlabel("Education")
plt.ylabel("Average Spending")
plt
```

```
<module 'matplotlib.pyplot' from '/usr/local/lib/python3.13/dist-packages/matplotlib/pyplot.py'>
```



```
df["AcceptedAny"]=df[["AcceptedCmp1","AcceptedCmp2","AcceptedCmp3","AcceptedCmp4","AcceptedCmp5","Response"]].sum(axis=1)
df["AcceptedAny"].unique()
```

```
array([1, 0, 3, 2, 4, 5])
```

```
df["AcceptedAny"]=df["AcceptedAny"].apply(lambda x: 1 if x>=1 else 0)
df["AcceptedAny"].unique()
```

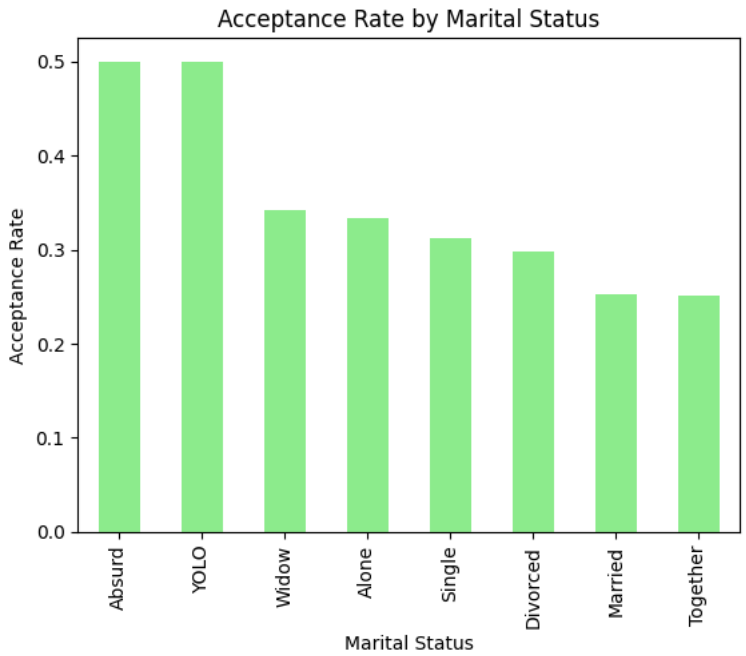
```
array([1, 0])
```

```
group2 = df.groupby("Marital_Status")["AcceptedAny"].mean().sort_values(ascending=False)
group2
```

Marital_Status	AcceptedAny
Absurd	0.500000
YOLO	0.500000
Widow	0.342105
Alone	0.333333
Single	0.312102
Divorced	0.297414
Married	0.252042
Together	0.251309

dtype: float64

```
group2.plot(kind="bar", color="lightgreen", title="Acceptance Rate by Marital Status")
plt.xlabel("Marital Status")
plt.ylabel("Acceptance Rate")
plt.show()
```




```
bins = [18, 30, 40, 50, 60, 70, 90]
labels = ["18-29", "30-39", "40-49", "50-59", "60-69", "70+"]
```

```
df["AgeGroup"] = pd.cut(df["Age"], bins=bins, labels=labels)
```

```
df["AgeGroup"]
```

	AgeGroup
0	60-69
1	70+
2	50-59
3	40-49
4	40-49
...	...
2235	50-59
2236	70+
2237	40-49
2238	60-69
2239	70+

2216 rows × 1 columns

dtype: category

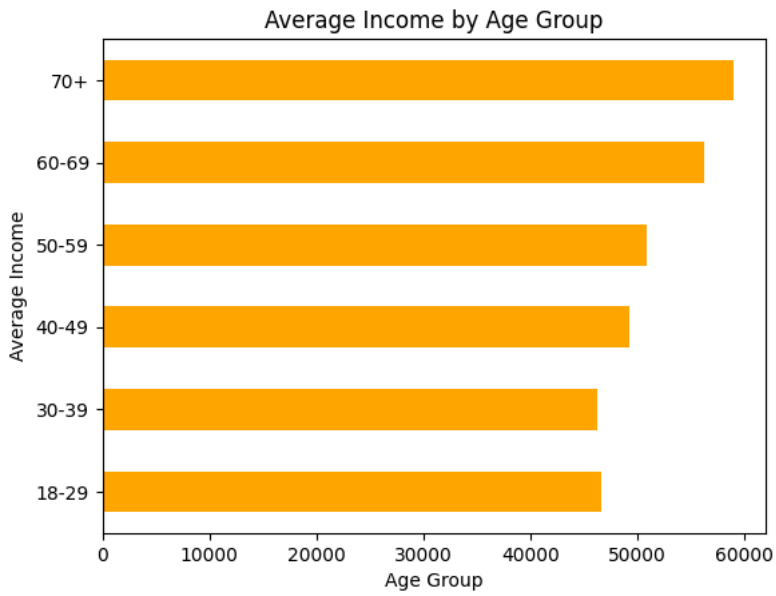
```
group3 = df.groupby("AgeGroup")["Income"].mean()
group3
```

```
/tmp/ipykernel_1661/197233607.py:1: FutureWarning: The default of observed=False is deprecated and will be changed to True i
  group3 = df.groupby("AgeGroup")["Income"].mean()
```

AgeGroup	Income
18-29	46658.000000
30-39	46283.028302
40-49	49224.877034
50-59	50812.913303
60-69	56200.827887
70+	58944.316294

dtype: float64

```
group3.plot(kind="barh", color="orange", title="Average Income by Age Group")
plt.xlabel("Age Group")
plt.ylabel("Average Income")
plt.show()
```




```
df.head()
```

	ID	Year_Birth	Education	Marital_Status	Income	Kidhome	Teenhome	Dt_Customer	Recency	MntWines	...	Complain	Z_
0	5524	1957	Graduation	Single	58138.0	0	0	2012-09-04	58	635	...	0	
1	2174	1954	Graduation	Single	46344.0	1	1	2014-03-08	38	11	...	0	
2	4141	1965	Graduation	Together	71613.0	0	0	2013-08-21	26	426	...	0	
3	6182	1984	Graduation	Together	26646.0	1	0	2014-02-10	26	11	...	0	
4	5324	1981	PhD	Married	58293.0	1	0	2014-01-19	94	173	...	0	

5 rows × 35 columns

```
df.columns
```

```
Index(['ID', 'Year_Birth', 'Education', 'Marital_Status', 'Income', 'Kidhome',
       'Teenhome', 'Dt_Customer', 'Recency', 'MntWines', 'MntFruits',
       'MntMeatProducts', 'MntFishProducts', 'MntSweetProducts',
       'MntGoldProds', 'NumDealsPurchases', 'NumWebPurchases',
       'NumCatalogPurchases', 'NumStorePurchases', 'NumWebVisitsMonth',
       'AcceptedCmp3', 'AcceptedCmp4', 'AcceptedCmp5', 'AcceptedCmp1',
       'AcceptedCmp2', 'Complain', 'Z_CostContact', 'Z_Revenue', 'Response',
       'Age', 'Total_Children', 'Total_Spend', 'Customer_Since', 'AcceptedAny',
       'AgeGroup'],
      dtype='object')
```

"Age", "Income", "Total_Spend", "NumWebPurchases", "NumStorePurchases", "NumWebVisitsMonth", "Recency"]

```
features = ["Age", "Income", "Total_Spend", "NumWebPurchases", "NumStorePurchases", "NumWebVisitsMonth", "Recency"]
```

```
X = df[features].copy()
```

```
X
```

	Age	Income	Total_Spend	NumWebPurchases	NumStorePurchases	NumWebVisitsMonth	Recency
0	68	58138.0	1617	8	4	7	58
1	71	46344.0	27	1	2	5	38
2	60	71613.0	776	8	10	4	26
3	41	26646.0	53	2	4	6	26
4	44	58293.0	422	5	6	5	94
...	...	...	...	...	...	...	...
2235	58	61223.0	1341	9	4	5	46
2236	79	64014.0	444	8	5	7	56
2237	44	56981.0	1241	2	13	6	91
2238	69	69245.0	843	6	10	3	8
2239	71	52869.0	172	3	4	7	40

2216 rows × 7 columns

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
```

```
X_scaled = scaler.fit_transform(X)
```

```
X_scaled
```

```
array([[ 0.98644293,  0.2340627 ,  1.67548812, ..., -0.55414289,
         0.69323197,  0.31053212],
       [ 1.23680074, -0.23455948, -0.96235832, ..., -1.16951781,
        -0.1315745 , -0.38050944],
       [ 0.31882209,  0.76947764,  0.28024985, ...,  1.29198186,
        -0.54397773, -0.79513438],
       ...,
       [-1.01641959,  0.18809052,  1.05169551, ...,  2.21504423,
         0.28082874,  1.4507507 ],
       [ 1.06989553,  0.67538765,  0.39140438, ...,  1.29198186,
        -0.95638097, -1.41707178],
       [ 1.23680074,  0.02470453, -0.7218    , ..., -0.55414289,
         0.69323197, -0.31140528]], shape=(2216, 7))
```

```
from sklearn.cluster import KMeans
```

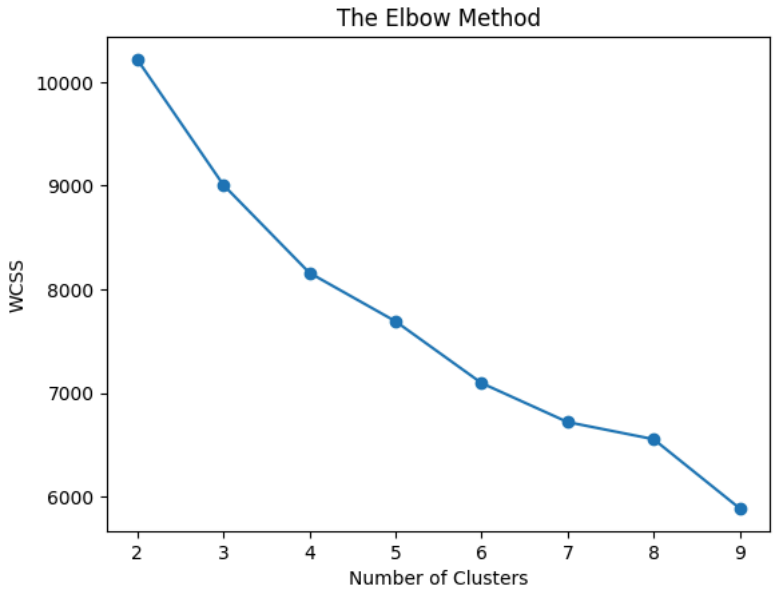
```
wcss =[]
```

```
for i in range(2,10):
     kmeans = KMeans(n_clusters=i)
     kmeans.fit(X_scaled)
     wcss.append(kmeans.inertia_)
```

```
wcss
```

```
[10218.620143748776,
 9007.783808892209,
 8159.023320597261,
 7691.207776838514,
 7094.7501087969285,
 6718.186566467726,
 6552.69512706076,
 5885.524789251269]
```

```
plt.plot(range(2,10),wcss,marker = "o")
plt.title("The Elbow Method")
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.show()
```




```
kmeans = KMeans(n_clusters =6)
df["Clusters"] = kmeans.fit_predict(X_scaled)
```

```
df["Clusters"]
```

	Clusters
0	2
1	1
2	2
3	1
4	3
...	...
2235	2
2236	2
2237	4
2238	5
2239	1

2216 rows × 1 columns

dtype: int32

```
df.head()
```

	ID	Year_Birth	Education	Marital_Status	Income	Kidhome	Teenhome	Dt_Customer	Recency	MntWines	...	Z_CostContac
0	5524	1957	Graduation	Single	58138.0	0	0	2012-09-04	58	635	...	
1	2174	1954	Graduation	Single	46344.0	1	1	2014-03-08	38	11	...	
2	4141	1965	Graduation	Together	71613.0	0	0	2013-08-21	26	426	...	
3	6182	1984	Graduation	Together	26646.0	1	0	2014-02-10	26	11	...	
4	5324	1981	PhD	Married	58293.0	1	0	2014-01-19	94	173	...	

5 rows × 36 columns

```
cluster_summery = df.groupby("Clusters")[features].mean()
```

```
cluster_summery
```

Clusters	Age	Income	Total_Spend	NumWebPurchases	NumStorePurchases	NumWebVisitsMonth	Recency
0	55.222222	5165.777778	88.555556	0.333333	0.222222	17.333333	43.333333
1	52.369643	34773.541071	120.392857	2.237500	3.389286	6.414286	24.098214
2	59.752969	59512.230404	879.622328	7.961995	7.790974	6.437055	46.745843
3	54.959559	36548.547794	135.277574	2.422794	3.487132	6.375000	74.963235
4	46.601719	78013.163324	1301.581662	4.684814	8.681948	2.724928	48.939828
5	70.126126	72362.105105	1137.831832	4.480480	8.252252	2.732733	51.612613

```
df["Clusters"].value_counts()
```

Clusters	count
1	560
3	544
2	421
4	349
5	333
0	9

dtype: int64

```
from sklearn.decomposition import PCA
pca = PCA(n_components=2)
```

```
pca_data = pca.fit_transform(X_scaled)
```

```
pca_data
```

```
array([[ 1.1075188 , -0.21175951],
       [-1.33673385,  0.269839  ],
       [ 1.88227676, -1.01416098],
       ...,
       [ 1.1535966 ,  1.15056999],
       [ 1.88768024, -1.16073698],
       [-0.84182091, -0.15963623]], shape=(2216, 2))
```

```
df["PCA1"]
```